

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criteria	Relevance to Application (1.25)	Use of Cloud Concepts (1.25)	Practical Insight (1)	Completeness (0.75)	Clarity (0.75)	TOTAL (5)
Weightage	25%	25%	20%	15%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Application Based Questions

1. How does cloud federation improve resource sharing and availability across datacenters?
2. What are the key security issues in virtualization, and how can they be mitigated?
3. How does a software-defined datacenter enhance scalability and management?
4. How can identity and access control ensure data privacy in cloud-based healthcare systems?
5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

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COURSE NAME: CLOUD COMPUTING AND BIG DATA ANALYSIS

Q1. How does cloud federation improve resource sharing and availability across datacenters?

Answer:

Cloud federation is the collaboration of multiple cloud providers or datacenters to share computing resources. It allows workloads to move between clouds when one datacenter becomes overloaded or unavailable.

Benefits:

- Improves resource sharing among multiple datacenters.
- Increases system availability and reliability.
- Supports load balancing across clouds.
- Reduces service downtime during failures.
- Provides better disaster recovery and business continuity.

Conclusion:

Cloud federation enhances resource utilization, ensures high availability, and improves overall cloud performance.

Q2. What are the key security issues in virtualization, and how can they be mitigated?

Answer:

Virtualization introduces several security risks because multiple virtual machines share the same physical hardware.

Security Issues:

- VM escape attacks.
- Hypervisor vulnerabilities.
- Unauthorized access to virtual machines.
- Data leakage between VMs.
- Malware attacks.

Mitigation Methods:

- Use secure and updated hypervisors.
- Apply Multi-Factor Authentication (MFA).
- Encrypt virtual machine data.

- Use firewalls and network segmentation.
- Regularly patch and monitor the virtual environment.

Conclusion:

Implementing strong security measures protects virtualized environments from cyber threats and unauthorized access.

Q3. How does a software-defined datacenter enhance scalability and management?

Answer:

A Software-Defined Datacenter (SDDC) virtualizes compute, storage, and networking resources and manages them through software.

Advantages:

- Automatically allocates resources based on demand.
- Enables rapid scaling of applications.
- Simplifies infrastructure management.
- Reduces operational costs.
- Improves resource utilization and flexibility.

Conclusion:

An SDDC provides efficient resource management, higher scalability, and simplified cloud operations.

Q4. How can identity and access control ensure data privacy in cloud-based healthcare systems?

Answer:

Identity and Access Control (IAC) ensures that only authorized healthcare professionals can access patient information.

Security Measures:

- Role-Based Access Control (RBAC).
- Multi-Factor Authentication (MFA).
- Single Sign-On (SSO).
- Data encryption during storage and transmission.
- Audit logs for monitoring user activities.
- Regular access reviews and permission updates.

Conclusion:

Identity and access control protect patient privacy, prevent unauthorized access, and ensure compliance with healthcare regulations.

Q5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

Answer:

Virtualization allows multiple virtual machines to run on a single physical server, while a hypervisor manages these virtual machines efficiently.

Benefits:

- Supports rapid creation of virtual machines.
- Improves CPU, memory, and storage utilization.
- Enables dynamic resource allocation.
- Reduces hardware costs.
- Provides high scalability and flexibility.
- Supports live migration of virtual machines without downtime.

Conclusion:

Virtualization and hypervisors improve cloud scalability, optimize resource usage, and reduce infrastructure costs.